INTEGRAL INDEFINIDA

Dada uma função $F(x)$ é chamada PRIMITIVA da função $f(x)$ em um intervalo I se, para todo $x \in I$, temos $F'(x) = f(x)$

EX01 $F(x) = \frac{x^3}{3}$ é uma PRIMITIVA de $f(x) = x^2$ $\quad F'(x) = \frac{3x^2}{3} = x^2$ $\quad \frac{d}{dx}$

EX02 $F(x) = \frac{1}{2}\,\text{sen}(2x)$ é uma PRIMITIVA de $f(x) = \cos(2x)$ $\quad F'(x) = \frac{1}{2}\cdot\cos(2x)\cdot 2 = \cos(2x)$

EX03 $F(x) = 5x^2 + 6x + 2$ é uma PRIMITIVA de $f(x) = 10x + 6$ $\quad F'(x) = 10x + 6$

$F(x) = 5x^2 + 6x + 3 \rightarrow$ PRIMITIVA $\quad f(x) = 10x + 6$

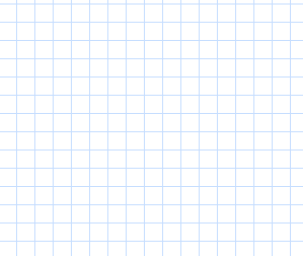
Então, se $F(x)$ é uma primitiva de $f(x)$, a expressão $F(x)+c$ é chamada INTEGRAL INDEFINIDA da função $f(x)$ sendo escrita por:

$$\boxed{\int f(x)\,dx = F(x) + c}$$

- $\int \rightarrow$ sinal de integração
- $dx \rightarrow$ indica a variável de integração
- $c \rightarrow$ constante

$\int 10x + 6\,dx = 5x^2 + 6x + c$

$\int 10x\,dx + \int 6\,dx = \frac{10x^2}{2} + \frac{6x^1}{1} + c = 5x^2 + 6x + c$



$\int_a^b f(x)\,dx \rightarrow$ INTEGRAL DEFINIDA

PROPRIEDADES

i) $\int k\,f(x)\,dx = k\int f(x)\,dx \quad \forall\, k \in \mathbb{R}$

ii) $\int f(x) \pm g(x)\,dx = \int f(x)\,dx \pm \int g(x)\,dx$

PÁG 245 TABELA DE INTEGRAIS IMEDIATAS

1) $\int u^m\,du = \frac{u^{m+1}}{m+1} + c \quad m \neq -1$

2) $\int \frac{du}{u} = \int u^{-1}\,du = \ln|u| + c$

3) $\int a^u\,du = \frac{a^u}{\ln(a)} + c \quad , a > 0$ e $a \neq 1$

4) $\int e^u\,du = e^u + c$

5) $\int \text{sen}(u)\,du = -\cos(u) + c$ $\quad F(x) = -\cos(x)$; $F'(x) = +\text{sen}(x)$

6) $\int \cos(u)\,du = \text{sen}(u) + c$

7) $\int \sec^2(u)\,du = \text{tg}(u) + c$

⋮

$\int x^2\,dx = \frac{x^{2+1}}{2+1} + c = \frac{x^3}{3} + c$

$F(x) = \frac{x^3}{3} + c \rightarrow F'(x) = f(x) = \frac{3x^2}{3} = x^2$

$F(x) = \frac{x^3}{3} + 5 \rightarrow F'(x) = f(x) = \frac{3x^2}{3} + 0 = x^2$

CALCULE AS INTEGRAIS INDEFINIDAS

EX01 $f(x) = 2x + 5$

$\int 2x + 5\,dx = \int 2x\,dx + \int 5\,dx = 2\int x\,dx + 5\int 1\,dx = 2\cdot\frac{x^2}{2} + 5\frac{x^1}{1} + c = x^2 + 5x + c$

$F(x) = x^2 + 5x + c$; $F'(x) = 2x + 5$

EX02 $f(x) = \sqrt{x}$

$\int \sqrt{x}\,dx = \int x^{1/2}\,dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + c = \frac{x^{3/2}}{3/2} + c = \frac{2}{3}x^{3/2} + c = \frac{2}{3}\sqrt{x^3} + c$

EX03 $f(x) = \frac{7}{x^3}$

$\int \frac{7}{x^3}\,dx = \int 7x^{-3}\,dx = 7\int x^{-3}\,dx = \frac{7x^{-2}}{-2} + c = -\frac{7}{2}x^{-2} + c = -\frac{7}{2x^2} + c$

$F(x) = -\frac{7}{2}x^{-2} + c$; $F'(x) = \frac{7\cdot 2}{2}x^{-3} + 0$; $\boxed{F'(x) = 7x^{-3}}$

EX04 $f(x) = 5^x$ $\quad \int 5^x\,dx = \frac{5^x}{\ln(5)} + c$

EX05 $f(x) = \frac{7}{x}$ $\quad \int \frac{7}{x}\,dx = \int 7\cdot x^{-1}\,dx = 7\ln|x| + c$

$F(x) = 7\ln|x| + c$; $F'(x) = 7\cdot\frac{1}{x} + 0 = \frac{7}{x}$

EX06 $f(x) = \frac{2}{3}x^3 - 3x + 1$

$\int \frac{2}{3}x^3 - 3x + 1\,dx = \frac{2}{3}\int x^3\,dx - 3\int x\,dx + \int 1\,dx$
$= \frac{2}{3}\cdot\frac{x^4}{4} - \frac{3x^2}{2} + x + c$
$= \frac{x^4}{6} - \frac{3}{2}x^2 + x + c$

TÉCNICAS DE INTEGRAÇÃO

INTEGRAÇÃO POR SUBSTITUIÇÃO

Decorre da regra da cadeia, de forma que:

$$\int f(g(x))\cdot g'(x)\,dx = F(g(x)) + c$$

Definindo $u = g(x)$, então $\frac{du}{dx} = g'(x) \rightarrow du = g'(x)dx$, logo:

$$\boxed{\int f(u)\,du = F(u) + c}$$

$\int x^2\,dx = \frac{x^3}{3} + c \rightarrow \int (x+1)^2\,dx = ?$

$u = x+1 \quad \frac{du}{dx} = 1 \rightarrow du = 1dx$

$\int u^2\,du = \frac{u^3}{3} + c \rightarrow \boxed{\frac{(x+1)^3}{3} + c}$

$\int e^x\,dx = e^x + c \rightarrow \int e^{4x}\,dx = ?$

$u = 4x \quad \frac{du}{dx} = 4 \rightarrow du = 4dx \rightarrow dx = \frac{du}{4}$

$\int e^{4x}\,dx = \int e^u\frac{du}{4} = \frac{1}{4}\int e^u\,du = \frac{1}{4}e^u + c \rightarrow \boxed{\frac{1}{4}e^{4x} + c}$

$\int \text{sen}(x)\,dx = -\cos(x) + c \rightarrow \int \text{sen}(4x)\,dx = ?$

$u = 4x \quad du = 4dx \rightarrow \frac{du}{4} = dx$

$\int \text{sen}(4x)\,dx = \int \text{sen}(u)\frac{du}{4} = \frac{1}{4}\int \text{sen}(u)\,du = -\frac{1}{4}\cos(u) + c = \boxed{-\frac{1}{4}\cos(4x) + c}$

$\int \text{tg}(x)\,dx = -\ln|\cos(x)| + c$?

$\int \text{tg}\,x\,dx = \int \frac{\text{sen}\,x\,dx}{\cos x} = \int -\frac{du}{u}$

$u = \cos(x) \quad \frac{du}{dx} = -\text{sen}(x) \rightarrow du = -\text{sen}(x)\,dx$

$= -\int \frac{du}{u}$
$= -\ln|u| + c$

$\boxed{\int \text{tg}(x)\,dx = -\ln|\cos(x)| + c}$

CALCULE AS INTEGRAIS:

EX01 $\int (x^3 + 7x - 1)^2\cdot\underbrace{(3x^2 + 7)\,dx}_{du}$

$u = x^3 + 7x - 1 \quad du = (3x^2 + 7)dx$

$\int u^2\,du = \frac{u^3}{3} + c = \boxed{\frac{(x^3 + 7x - 1)^3}{3} + c}$

EX02 $\int \sqrt{5x+1}\,dx$

$u = 5x + 1$
$du = 5dx \rightarrow \frac{du}{5} = dx$

$\int \sqrt{u}\,\frac{du}{5} = \frac{1}{5}\int u^{1/2}\,du = \frac{1}{5}\cdot\frac{u^{3/2}}{3/2} + c = \frac{1}{5}\cdot\frac{2}{3}\sqrt{u^3} + c = \boxed{\frac{2}{15}\sqrt{(5x+1)^3} + c}$

EX03 $\int 16x\cdot\cos(8x^2 + 6)\,dx = \int \cos(u)\,du = \text{sen}(u) + c = \boxed{\text{sen}(8x^2 + 6) + c}$

$u = 8x^2 + 6$
$du = 16x\,dx$

EX04 $\int x^2 e^{5x^3}\,dx = \int e^u\frac{du}{15} = \frac{1}{15}\int e^u\,du = \frac{1}{15}e^u + c = \boxed{\frac{1}{15}e^{5x^3} + c}$

$u = 5x^3$
$du = 15x^2\,dx$
$\frac{du}{15} = x^2\,dx$

EX05 $\int \text{sen}^2(x)\cdot\cos(x)\,dx = \int u^2\,du = \frac{u^3}{3} + c = \boxed{\frac{\text{sen}^3(x)}{3} + c}$

$u = \text{sen}(x)$
$du = \cos(x)\,dx$

Prova $F(x) = \frac{\text{sen}^3(x)}{3} + c$; $F'(x) = \frac{3\,\text{sen}^2(x)}{3}\cdot\cos(x)$

EX06 $\int 3x\sqrt{1+x^2}\,dx = 3\int \sqrt{u}\,\frac{du}{2} = \frac{3}{2}\int u^{1/2}\,du = \frac{3}{2}\cdot\frac{u^{3/2}}{3/2} + c = \frac{3}{2}\cdot\frac{2}{3}\sqrt{u^3} + c = \boxed{\sqrt{(1+x^2)^3} + c}$

$u = 1 + x^2$
$du = 2x\,dx$
$\frac{du}{2} = x\,dx$

$\int e^{2x}\,\text{sen}(x)\,dx$
$\int 2x\,\ln(x+1)\,dx$ } INTEGRAÇÃO POR PARTES

RESOLVA AS INTEGRAIS POR SUBSTITUIÇÃO

EX01 $\int \text{tg}(3x)\,dx$

EX02 $\int 6x\cdot\sqrt{-x^2 + 1}\,dx$

EX03 $\int (x^2 + 4)\,3^{x^3 + 12x}\,dx$

EX04 $\int \frac{x}{3}\cdot\cos(x^2 + 1)\,dx$

EX05 $\int \frac{\ln x}{x}\,dx$

EX06 $\int \frac{e^x}{e^{2x} + 16}\,dx$

INTEGRAÇÃO POR PARTES

Sejam $f(x)$ e $g(x)$ duas funções deriváveis no intervalo I, temos:

$[f(x)\cdot g(x)]' = f'(x)\cdot g(x) + f(x)\cdot g'(x)$

$\int f(x)\,g'(x)\,dx = \int [f(x)\cdot g(x)]'\,dx - \int f'(x)\cdot g(x)\,dx$

$\int f(x)\,g'(x)\,dx = f(x)\cdot g(x) - \int f'(x)\cdot g(x)\,dx$

$f(x) = u \quad f'(x) = du$
$g(x) = v \quad g'(x) = dv$

$\boxed{\int u\,dv = uv - \int v\,du}$ INTEGRAL POR PARTES

EX01 $\int \underbrace{5x}_{u}\,\underbrace{e^{-2x}\,dx}_{dv}$

$u = 5x \quad du = 5dx$
$\int dv = \int e^{-2x}\,dx \rightarrow v = \frac{e^{-2x}}{-2} + c$

$\int u\,dv = u\cdot v - \int v\,du$ (Escolha mais simples)

$\int 5x\,e^{-2x}\,dx = 5x\cdot\frac{e^{-2x}}{-2} - \int \frac{e^{-2x}}{-2}\,5\,dx$
$= -\frac{5}{2}x\,e^{-2x} + \frac{5}{2}\int e^{-2x}\,dx$
$= -\frac{5}{2}x\,e^{-2x} - \frac{5}{2}\cdot\frac{e^{-2x}}{2} + c$

$\boxed{\int 5x\,e^{-2x}\,dx = -\frac{5}{2}x\,e^{-2x} - \frac{5}{4}e^{-2x} + c}$

EX02 $\int x^2\,\text{sen}(2x)\,dx$

$u = x^2 \quad du = 2x\,dx$
$\int dv = \int \text{sen}(2x)\,dx \quad v = -\frac{\cos(2x)}{2}$

INT. POR SUBSTITUIÇÃO: $\int \text{sen}(2x)\,dx$; $u = 2x$, $du = 2dx$; $\int \text{sen}(u)\frac{du}{2} = \frac{1}{2}\int \text{sen}(u)\,du = -\frac{1}{2}\cos(2x) + c$

$\int u\,dv = u\cdot v - \int v\,du$

$\int x^2\,\text{sen}(2x)\,dx = x^2\cdot\frac{-\cos(2x)}{2} - \int \frac{-\cos(2x)}{2}\,2x\,dx$
$= -\frac{x^2\cos(2x)}{2} + \int x\cdot\cos(2x)\,dx$ (INT. POR PARTES)
$= -\frac{x^2\cos(2x)}{2} + \left(\frac{x\,\text{sen}(2x)}{2} + \frac{\cos(2x)}{4}\right) + c$

$\boxed{\int x^2\,\text{sen}(2x)\,dx = -\frac{x^2\cos(2x)}{2} + \frac{x\cdot\text{sen}(2x)}{2} + \frac{\cos(2x)}{4} + c}$ Solução

$\int x\cos(2x)\,dx$; $u = x \quad du = 1dx$; $\int dv = \int \cos(2x)\,dx \quad v = \frac{\text{sen}(2x)}{2}$

$\int x\cdot\cos(2x)\,dx = \frac{x\cdot\text{sen}(2x)}{2} - \int \frac{\text{sen}(2x)}{2}\,dx$
$= \frac{x\cdot\text{sen}(2x)}{2} - \frac{1}{2}\int \text{sen}(2x)\,dx$
$= \frac{x\cdot\text{sen}(2x)}{2} - \frac{1}{2}\cdot\frac{-\cos(2x)}{2}$
$= \frac{x\cdot\text{sen}(2x)}{2} + \frac{\cos(2x)}{4} + c$

EX03 $\int \sqrt{x}\,\ln(x)\,dx = \int x^{1/2}\,\ln(x)\,dx$

$u = \ln|x| \quad \frac{du}{dx} = \frac{1}{x} \rightarrow du = \frac{1}{x}dx$
$\int dv = \int x^{1/2}\,dx \quad v = \frac{x^{3/2}}{3/2} \rightarrow v = \frac{2}{3}x^{3/2}$

$u\cdot v - \int v\,du$

$\int x^{1/2}\ln(x)\,dx = \ln(x)\cdot\frac{2}{3}x^{3/2} - \int \frac{2}{3}x^{3/2}\cdot\frac{1}{x}\,dx$ (Simplificar: $x^{3/2}\cdot x^{-1} = x^{1/2}$)
$= \frac{2}{3}x^{3/2}\ln(x) - \frac{2}{3}\int x^{1/2}\,dx$
$= \frac{2}{3}\sqrt{x^3}\ln(x) - \frac{2}{3}\cdot\frac{2}{3}\sqrt{x^3} + c$

$\boxed{\int \sqrt{x}\ln(x)\,dx = \frac{2}{3}\sqrt{x^3}\ln(x) - \frac{4}{9}\sqrt{x^3} + c}$

$\int x^m\,dx \rightarrow \frac{x^{m+1}}{m+1} + c$

$\int x^{1/2}\,dx = \frac{x^{3/2}}{3/2} + c = \frac{2x^{3/2}}{3} + c = \frac{2}{3}x^{3/2}$

$\int \ln(x)\,dx$; $u = \ln|x| \quad du = \frac{1}{x}dx$; $\int dv = \int dx \quad v = x$

$= \ln x\cdot x - \int x\cdot\frac{1}{x}\,dx$

$\boxed{\int \ln(x)\,dx = x\cdot\ln(x) - x}$

EX04 $\int e^{3x}\cos(x)\,dx$

$u = \cos(x) \quad du = -\text{sen}(x)\,dx$
$\int dv = \int e^{3x}\,dx \quad v = \frac{e^{3x}}{3}$

$\int e^{3x}\cos(x)\,dx = \cos(x)\cdot\frac{e^{3x}}{3} - \int \frac{e^{3x}}{3}\cdot(-\text{sen}(x))\,dx$ (não houve simplificação)
$= \frac{e^{3x}\cos(x)}{3} + \frac{1}{3}\int e^{3x}\,\text{sen}(x)\,dx$

$\int e^{3x}\cos(x)\,dx = \frac{e^{3x}\cos(x)}{3} + \frac{1}{3}\left(\frac{e^{3x}\,\text{sen}(x)}{3} - \frac{1}{3}\int e^{3x}\cos(x)\,dx\right)$

$\int e^{3x}\cos(x)\,dx = \frac{e^{3x}\cos(x)}{3} + \frac{e^{3x}\,\text{sen}(x)}{9} - \frac{1}{9}\int e^{3x}\cos(x)\,dx$

$1\int e^{3x}\cos(x)\,dx + \frac{1}{9}\int e^{3x}\cos(x)\,dx = \frac{e^{3x}\cos(x)}{3} + \frac{e^{3x}\,\text{sen}(x)}{9} + c$

$\frac{10}{9}\int e^{3x}\cos(x)\,dx = \frac{e^{3x}\cos(x)}{3} + \frac{e^{3x}\,\text{sen}(x)}{9} + c$

$\boxed{\int e^{3x}\cos(x)\,dx = \frac{9}{10}\left(\frac{e^{3x}\cos(x)}{3} + \frac{e^{3x}\,\text{sen}(x)}{9}\right) + c}$

$\int e^{3x}\,\text{sen}(x)\,dx$; $u = \text{sen}(x) \quad du = \cos(x)\,dx$; $dv = e^{3x}\,dx \quad v = \frac{e^{3x}}{3} + c$

$\int e^{3x}\,\text{sen}(x)\,dx = \text{sen}(x)\,\frac{e^{3x}}{3} - \int \frac{e^{3x}}{3}\cos(x)\,dx$
$= \frac{e^{3x}}{3}\,\text{sen}(x) - \frac{1}{3}\int e^{3x}\cos(x)\,dx$

PROVA $\int \ln(x)\,dx = x\,\ln(x) - x + c$

$F(x) = x\,\ln(x) - x + c$
$F'(x) = 1\cdot\ln(x) + x\cdot\frac{1}{x} - 1 + 0$
$F'(x) = \ln(x)$

INTEGRAÇÃO DE FUNÇÕES RACIONAIS POR FRAÇÕES PARCIAIS

A decomposição em função racional $\left(f(x) = \frac{P(x)}{Q(x)}\right)$ em frações mais simples são denominadas de frações parciais de modo que o denominador $Q(x)$ seja decomposto em fatores lineares e/ou quadráticos irredutíveis.

CASO 1: Os fatores de $Q(x)$ são lineares e distintos $\quad f(x) = \frac{A}{x - x'} + \frac{B}{x - x''}$

EX01 $\int \frac{x - 2}{x^3 - 4x^2 - 5x}\,dx$

$g(x) = x^3 - 4x^2 - 5x = 0$ Decomposição do polinômio
$x(x^2 - 4x - 5) = 0$
$x' = 0 \rightarrow (x - 0) = 0$
$x^2 - 4x - 5 = 0$
$x'' = 5 \rightarrow (x - 5) = 0$
$x''' = -1 \rightarrow (x + 1) = 0$

$x^3 - 4x^2 - 5x = x\cdot(x-5)\cdot(x+1)$ fatores lineares e distintos

frações parciais: $\frac{x - 2}{x^3 - 4x^2 - 5x} = \frac{A}{x - 0} + \frac{B}{x - 5} + \frac{C}{x + 1}$

$\frac{x - 2}{x^3 - 4x^2 - 5x} = \frac{A(x-5)(x+1) + Bx(x+1) + Cx(x-5)}{x\cdot(x-5)\cdot(x+1)}$

$x - 2 = A(x^2 - 4x - 5) + Bx^2 + Bx + Cx^2 - 5Cx$

$x - 2 = Ax^2 + Bx^2 + Cx^2 - 4Ax + Bx - 5Cx - 5A$

$1x - 2 = x^2(A + B + C) + x(-4A + B - 5C) - 5A$

$\begin{cases} A + B + C = 0 \\ -4A + B - 5C = 1 \\ -5A = -2 \end{cases}$ $\quad \boxed{A = 2/5}$

$\begin{cases} B + C = -2/5 \\ -4\cdot\frac{2}{5} + B - 5C = 1 \end{cases}$

$\begin{cases} B + C = -2/5 \\ B - 5C = 1 + 8/5 \end{cases} \rightarrow \begin{cases} -B - C = +2/5 \\ B - 5C = 13/5 \end{cases}$ (-1)

$-6C = 3$
$\boxed{C = -1/2}$

$B + C = -2/5$
$B = -2/5 + 1/2$
$B = \frac{-4 + 5}{10}$
$\boxed{B = 1/10}$

função racional → frações parciais

$\int \frac{x - 2}{x^3 - 4x^2 - 5x}\,dx = \int \frac{2/5}{x} + \frac{1/10}{x - 5} - \frac{1/2}{x + 1}\,dx$
$= \int \frac{2/5}{x}\,dx + \int \frac{1/10}{x - 5}\,dx - \int \frac{1/2}{x + 1}\,dx$

$\frac{2}{5}\int \frac{dx}{x} = \frac{2}{5}\ln|x|$

$\frac{1}{10}\int \frac{dx}{x - 5}$; $u = x - 5 \quad du = 1dx$; $\frac{1}{10}\int \frac{du}{u} = \frac{1}{10}\ln|x - 5|$

$\frac{1}{2}\int \frac{dx}{x + 1}$; $u = x + 1 \quad du = 1dx$; $\frac{1}{2}\int \frac{du}{u} = \frac{1}{2}\ln|x + 1|$

$\boxed{\int \frac{x - 2}{x^3 - 4x^2 - 5x}\,dx = \frac{2}{5}\ln|x| + \frac{1}{10}\ln|x - 5| - \frac{1}{2}\ln|x + 1| + c}$

EX02 $\int \frac{3x^2 + 20x + 25}{x^2 + 7x + 10}\,dx \longrightarrow \int 3 + \frac{-x - 5}{x^2 + 7x + 10}\,dx = 3x + \int \frac{-x - 5}{x^2 + 7x + 10}\,dx$

DIVISÃO DE POLINÔMIOS

$3x^2 + 20x + 25 \,|\, x^2 + 7x + 10$
$-3x^2 - 21x - 30 \quad 3$
$-x - 5$

$\frac{3x^2 + 20x + 25}{x^2 + 7x + 10} = 3 + \frac{-x - 5}{x^2 + 7x + 10}$ frações parciais

Decomposição
$x^2 + 7x + 10 = 0$
$x' = -5 \rightarrow x + 5 = 0$
$x'' = -2 \rightarrow x + 2 = 0$ fatores lineares e distintos

$\frac{-x - 5}{x^2 + 7x + 10} = \frac{A}{x + 5} + \frac{B}{x + 2}$

$\frac{-x - 5}{x^2 + 7x + 10} = \frac{A(x+2) + B(x+5)}{(x+5)(x+2)}$

$-x - 5 = Ax + 2A + Bx + 5B$

$\begin{cases} A + B = -1 \\ 2A + 5B = -5 \end{cases}$ (-2)

$3B = -3$
$\boxed{B = -1}$

$A + B = -1$
$A = -1 + 1$
$\boxed{A = 0}$

$\int \frac{-x - 5}{x^2 + 7x + 10}\,dx = \int \frac{0}{x + 5} + \frac{-1}{x + 1}\,dx$
$= \int -\frac{1}{x + 1}\,dx$
$= -1\cdot\ln|x + 1| + c$

$u = x + 1 \quad du = 1dx \quad \int -\frac{1du}{u} = -1\ln|u|$

$\boxed{\int \frac{3x^2 + 20x + 25}{x^2 + 7x + 10}\,dx = 3x - 1\ln|x + 1| + c}$

CASO 2. Os fatores de $Q(x)$ são lineares e alguns deles se repetem $\quad f(x) = \frac{A}{(x - x')^m} + \frac{B}{(x - x')^{m-1}} \cdots \frac{C}{(x - x')^1}$

EX03 $\int \frac{1}{x^3 - 5x^2 + 8x - 4}\,dx$

Decomposição
$x^3 - 5x^2 + 8x - 4 = 0$
$x = 1 \rightarrow (x - 1)$

$x^3 - 5x^2 + 8x - 4 \,|\, x - 1$
$-x^3 + x^2 \quad x^2 - 4x + 4$
$-4x^2 + 8x$
$+4x^2 - 4x$
$4x - 4$
$-4x + 4$
0

$(x - 1)\cdot(x^2 - 4x + 4) = 0$
$x - 1 = 0 \quad \boxed{x' = 1}$
$x^2 - 4x + 4 = 0 \quad \boxed{x'' = 2} \rightarrow (x - 2) \quad \boxed{x''' = 2} \rightarrow (x - 2)$

fatores lineares e repetidos

$\frac{1}{x^3 - 5x^2 + 8x - 4} = \frac{A}{x - 1} + \frac{B}{(x - 2)^2} + \frac{C}{(x - 2)^1}$

$\frac{1}{x^3 - 5x^2 + 8x - 4} = \frac{A(x-2)^2 + B(x-1) + C(x-1)\cdot(x-2)}{(x-1)\cdot(x-2)^2}$

$1 = A(x^2 - 4x + 4) + Bx - B + C\cdot(x^2 - 3x + 2)$

$1 = Ax^2 - 4Ax + 4A + Bx - B + Cx^2 - 3Cx + 2C$

$\begin{cases} A + C = 0 \\ -4A + B - 3C = 0 \\ 4A - B + 2C = 1 \end{cases}$ +

$A + C = 0$
$-C = 1 \rightarrow \boxed{C = -1}$
$\boxed{A = 1}$

$4A - B + 2C = 1$
$4 - B - 2 = 1$
$-B = 1 - 2$
$\boxed{B = 1}$

$\int \frac{1}{x^3 - 5x^2 + 8x - 4}\,dx = \int \frac{1}{x - 1} + \frac{1}{(x - 2)^2} + \frac{-1}{x - 2}\,dx$
$= \int \frac{1}{x - 1}\,dx + \int \frac{1}{(x - 2)^2}\,dx - \int \frac{1}{x - 2}\,dx$

$u = x - 1 \quad du = 1dx \quad \int \frac{du}{u} = \ln|u| = \ln|x - 1|$

$u = x - 2 \quad du = 1dx \quad \int \frac{1du}{u^2} = \int u^{-2}\,du = \frac{u^{-1}}{-1} = -\frac{1}{u} = -\frac{1}{x - 2}$

$u = x - 2 \quad du = 1dx \quad \int \frac{du}{u} = \ln|u| = \ln|x - 2|$

$\boxed{\int \frac{1}{x^3 - 5x^2 + 8x - 4}\,dx = \ln|x - 1| - \frac{1}{x - 2} - \ln|x - 2| + c}$

EX04 $\int \frac{-2x + 1}{x^3 - 8x^2}\,dx$

CASO 3. Os fatores de $Q(x)$ são lineares e quadráticos irredutíveis, sendo que os fatores quadráticos não se repetem. $\quad f(x) = \frac{Cx + D}{ax^2 + bx + c}\ldots$

EX05 $\int \frac{x + 1}{x^2 + 6x + 12}\,dx$

Decomposição
$x^2 + 6x + 12 = 0$
$x = \frac{-6 \pm \sqrt{36 - 48}}{2}$
$x = \nexists\,\mathbb{R}$
quadrático irredutível

$\int \frac{x + 1}{x^2 + 6x + 12}\,dx$
$= \int \frac{x + 1}{x^2 + 6x + 9 + 3}\,dx$
$= \int \frac{x + 1}{(x + 3)^2 + 3}\,dx$

$u = x + 3 \rightarrow x = u - 3$
$du = 1dx$

$= \int \frac{u - 3 + 1}{u^2 + 3}\,du$
$= \int \frac{u - 2}{u^2 + 3}\,du$
$= \int \frac{u}{u^2 + 3}\,du - \int \frac{2}{u^2 + 3}\,du$
$= \frac{1}{2}\ln|u| - 2\cdot\int \frac{du}{u^2 + 3}$

$a^2 = 3 \quad a = \sqrt{3}$

$\int \frac{du}{a^2 + u^2} = \frac{1}{a}\cdot\text{arctg}\left(\frac{u}{a}\right) + c$

$\int \frac{u\,du}{u^2 + a^2} = \frac{1}{2}\ln|u| + c$

$\boxed{\int \frac{x + 1}{x^2 + 6x + 12}\,dx = \frac{1}{2}\ln|x + 3| - 2\cdot\frac{1}{\sqrt{3}}\cdot\text{arctg}\left(\frac{x + 3}{\sqrt{3}}\right) + c}$

EX06 $\int \frac{2x^2 + 5x + 4}{x^3 + x^2 + x - 3}\,dx$

$x^3 + x^2 + x - 3 = 0$
$\boxed{x = 1}$

$x^3 + x^2 + x - 3 \,|\, x - 1$
$-x^3 + x^2 \quad x^2 + 2x + 3$
$2x^2 + x$
$-2x^2 + 2x$
$3x - 3$
$-3x + 3$
0

$x^3 + x^2 + x - 3 = (x - 1)\cdot(x^2 + 2x + 3)$
$x - 1 = 0 \quad x = 1$
$x^2 + 2x + 3 = 0 \quad x \notin \mathbb{R}$ quadrático irredutível

$\frac{2x^2 + 5x + 4}{x^3 + x^2 + x - 3} = \frac{A}{x - 1} + \frac{Cx + D}{(x^2 + 2x + 3)}$

$\frac{2x^2 + 5x + 4}{x^3 + x^2 + x - 3} = \frac{A(x^2 + 2x + 3) + (Cx + D)\cdot(x - 1)}{(x - 1)\cdot(x^2 + 2x + 3)}$

$2x^2 + 5x + 4 = Ax^2 + 2Ax + 3A + Cx^2 - Cx + Dx - D$

$\begin{cases} A + C = 2 \\ 2A - C + D = 5 \\ 3A - D = 4 \end{cases}$ $\quad \boxed{C = 2 - A} \quad -D = 4 - 3A \quad \boxed{D = 3A - 4}$

$2A - (2 - A) + (3A - 4) = 5$
$2A - 2 + A + 3A - 4 = 5$
$6A = 11$
$A = 11/6$

$C = 2 - \frac{11}{6} \rightarrow \boxed{C = \frac{1}{6}}$

$D = 3\cdot\frac{11}{6} - 4 \rightarrow D = \frac{11}{2} - 4 \rightarrow \boxed{D = \frac{3}{2}}$

$\int \frac{2x^2 + 5x + 4}{x^3 + x^2 + x - 3}\,dx = \int \frac{11/6}{x - 1}\,dx + \int \frac{1/6\,x + 3/2}{x^2 + 2x + 3}\,dx$
$= \frac{11}{6}\ln|x - 1| + \int \frac{1/6\,x + 3/2}{x^2 + 2x + 1 + 2}\,dx$
$+ \int \frac{1/6\,x + 3/2}{(x + 1)^2 + 2}\,dx$

$u = x + 1 \quad x = u - 1$
$du = 1dx$

$= \int \frac{1/6\,(u - 1) + 3/2}{u^2 + 2}\,du$
$= \int \frac{1/6\,u - 1/6 + 3/2}{u^2 + 2}\,dx$
$= \int \frac{1/6\,u}{u^2 + 2}\,du + \int \frac{4/3}{u^2 + 2}\,du$ ← TABELA INTEGRAIS IMEDIATAS

$-\frac{1}{6} + \frac{3}{2} = \frac{-1 + 9}{6} = \frac{8}{6} = \frac{4}{3}$

$= \frac{1}{6}\cdot\frac{1}{2}\ln|u| + \frac{4}{3}\cdot\frac{1}{\sqrt{2}}\,\text{arctg}\left(\frac{u}{\sqrt{2}}\right) + c$

$\boxed{\int \frac{2x^2 + 5x + 4}{x^3 + x^2 + x - 3}\,dx = \frac{1}{12}\ln|x + 1| + \frac{4}{3\sqrt{2}}\,\text{arctg}\left(\frac{x + 1}{\sqrt{2}}\right) + c}$

EXERCÍCIOS INTEGRAÇÃO POR PARTES

EX01 $\int e^x\cos\frac{x}{2}\,dx$

EX02 $\int (x - 1)\cdot e^{-x}\,dx$

EX03 $\int x^2\,e^{5x}\,dx$

EX04 $\int x^2\cos(3x)\,dx$

EX05 $\int 5x^2\,\ln(x)\,dx$

EX06 $\int e^{3x}\cdot\cos(2x)\,dx$

EX07 $\int x\cdot\sqrt{x + 1}\,dx$

$\int x^2\,e^{5x}\,dx =$
$u = x^2 \quad du = 2x\,dx$
$\int dv = \int e^{5x}\,dx \quad v = \frac{e^{5x}}{5}$

$\int u\,dv = uv - \int v\,du$

$\int x^2 e^{5x}\,dx = x^2\frac{e^{5x}}{5} - \int \frac{e^{5x}}{5}\,2x\,dx \longrightarrow \int x^2 e^{5x}\,dx = \frac{x^2 e^{5x}}{5} - \frac{2}{5}\int x\,e^{5x}\,dx$ (2ª integração por partes)

$u = x \quad du = 1dx$
$\int dv = \int e^{5x}\,dx \quad v = \frac{e^{5x}}{5}$

$= \frac{x^2 e^{5x}}{5} - \frac{2}{5}\left(\frac{x\,e^{5x}}{5} - \frac{e^{5x}}{25} + c\right)$
$= \frac{x^2 e^{5x}}{5} - \frac{2x\,e^{5x}}{25} + \frac{2e^{5x}}{125} + c$

$\int e^{5x}\,dx = \int e^u\frac{du}{5} = \frac{e^u}{5} = \frac{e^{5x}}{5}$
$u = 5x \quad du = 5dx \quad \frac{du}{5} = dx$

$\int 5x^2\,\ln(x)\,dx$ (ALGÉBRICA; LOG → u)
$u = \ln(x) \quad du = \frac{1}{x}dx$
$\int dv = \int 5x^2\,dx \rightarrow v = \frac{5x^3}{3}$

$\int u\,dv = u\cdot v - \int v\,du$

$\int 5x^2\ln(x)\,dx = \ln(x)\cdot\frac{5x^3}{3} - \int \frac{5x^3}{3}\cdot\frac{1}{x}\,dx$
$= \frac{5}{3}\ln(x)\cdot x^3 - \frac{5}{3}\int x^2\,dx$
$= \frac{5}{3}\ln(x)\cdot x^3 - \frac{5}{3}\cdot\frac{x^3}{3} + c$
$= \frac{5}{3}\ln(x)\cdot x^3 - \frac{5x^3}{9} + c$

EXERCÍCIOS INTEGRAIS POR FRAÇÕES RACIONAIS

EX01 $\int \frac{x - 1}{x^3 + x^2 - 4x - 4}\,dx$

EX02 $\int \frac{1}{x^3 + 9x}\,dx$

EX03 $\int \frac{1}{x^3 + 8}\,dx$

EX04 $\int \frac{2x + 1}{2x^2 + 3x - 2}\,dx$

INTEGRAIS DO TIPO $\int \text{sen}^m x\,dx$ e $\int \cos^m x\,dx$, ONDE m É UM Nº INTEIRO POSITIVO

PARA $m \geq 2 \longrightarrow$ IDENTIDADES TRIGONOMÉTRICAS

$\text{sen}^2 x + \cos^2 x = 1 \longrightarrow$ se m for ímpar

$\text{sen}^2 x = \frac{1}{2} - \frac{1}{2}\cos(2x)$
$\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos(2x)$ } → se m for PAR

EX01 $\int \cos^3 x\,dx = \int \cos^2 x\cdot\cos x\,dx$
$= \int (1 - \text{sen}^2 x)\cdot\cos x\,dx$
$= \int \cos x\,dx - \int \text{sen}^2 x\cdot\cos x\,dx$

$\text{sen}^2 x + \cos^2 x = 1$
$\cos^2 x = 1 - \text{sen}^2 x$

$u = \text{sen}(x)$
$du = \cos(x)\,dx \quad \int u^2\,du = \frac{u^3}{3} + c$

$\boxed{\int \cos^3 x\,dx = -\text{sen}\,x - \frac{\text{sen}^3 x}{3} + c}$

EX02 $\int \text{sen}^4(2x)\,dx = \int \text{sen}^2(2x)\cdot\text{sen}^2(2x)\,dx$
$= \int \left(\frac{1}{2} - \frac{1}{2}\cos(4x)\right)\cdot\left(\frac{1}{2} - \frac{1}{2}\cos(4x)\right)dx$
$= \int \left(\frac{1}{4} - \frac{1}{2}\cos(4x) + \frac{1}{4}\cos^2(4x)\right)dx$
$= \int \frac{1}{4}dx - \int \frac{1}{2}\cos(4x)\,dx + \frac{1}{4}\int \left(\frac{1}{2} + \frac{1}{2}\cos(8x)\right)d$
$= \frac{1}{4}x - \frac{1}{2}\cdot\frac{\text{sen}(4x)}{4} + \frac{1}{4}\cdot\left(\frac{1}{2}x + \frac{1}{2}\frac{\text{sen}(8x)}{8}\right) + c$
$= \frac{1}{4}x + \frac{1}{8}x - \frac{\text{sen}(4x)}{8} + \frac{1}{64}\text{sen}(8x) + c$

$\text{sen}^2(2x) = \frac{1}{2} - \frac{1}{2}\cos(4x)$

$\cos^2(4x) = \frac{1}{2} + \frac{1}{2}\cos(8x)$

$\boxed{\int \text{sen}^4(2x)\,dx = \frac{3x}{8} - \frac{\text{sen}(4x)}{8} + \frac{\text{sen}(8x)}{64} + c}$

EX03 $\int \text{sen}^3 x\cdot\cos^2 x\,dx = \int \text{sen}^2 x\cdot\text{sen}\,x\cdot\cos^2 x\,dx$
$= \int (1 - \cos^2 x)\cdot\cos^2 x\cdot\text{sen}\,x\,dx$
$= \int \cos^2 x\cdot\text{sen}\,x\,dx - \int \cos^4 x\cdot\text{sen}(x)\,dx$

$\text{sen}^2 x = 1 - \cos^2 x$

$u = \cos x$
$du = -\text{sen}\,x\,dx$
$-du = \text{sen}(x)\,dx$
$\int -u^2\,du = -\frac{u^3}{3}$

$u = \cos(x)$
$du = -\text{sen}\,x\,dx$
$-du = \text{sen}\,x\,dx$
$\int -u^4\,du = -\frac{u^5}{5} + c$

$\boxed{\int \text{sen}^3 x\cdot\cos^2 x\,dx = -\frac{\cos^3 x}{3} - \frac{\cos^5 x}{5} + c}$

EX04 $\int \text{sen}(3x)\cdot\cos(2x)\,dx$

ADIÇÃO DE ARCOS
$\text{sen}\,A\cdot\cos B = \frac{1}{2}\left(\text{sen}(A+B) + \text{sen}(A-B)\right)$
$\text{sen}\,A\cdot\text{sen}\,B = \frac{1}{2}\left(\cos(A-B) - \cos(A+B)\right)$
$\cos A\cdot\cos B = \frac{1}{2}\left(\cos(A+B) + \cos(A-B)\right)$

$\text{sen}(3x)\cdot\cos(2x) = \frac{1}{2}\left(\text{sen}(5x) + \text{sen}(x)\right)$

$\int \text{sen}(3x)\cdot\cos(2x)\,dx = \int \frac{1}{2}\left(\text{sen}(5x) + \text{sen}(x)\right)dx$
$= \frac{1}{2}\left(\int \text{sen}(5x)\,dx + \int \text{sen}(x)\,dx\right)$
$= \frac{1}{2}\cdot\left(\frac{\cos(5x)}{5} + \cos(x) + c\right)$

$\boxed{\int \text{sen}(3x)\cdot\cos(2x)\,dx = \frac{\cos(5x)}{10} + \frac{\cos(x)}{2} + c}$

INTEGRAIS DO TIPO $\int \text{tg}^m x\,dx$ e $\int \text{cotg}^m x\,dx$ c/ m INTEIRO POSITIVO

$\text{tg}^2 x = \sec^2 x - 1 \quad$ e $\quad \text{cotg}^2 x = \text{cossec}^2 x - 1$

EX05 $\int \text{tg}^3(5x)\,dx = \int \text{tg}^2(5x)\cdot\text{tg}(5x)\,dx$
$= \int (\sec^2(5x) - 1)\cdot\text{tg}(5x)\,dx$
$= \int \sec^2(5x)\cdot\text{tg}(5x)\,dx - \int \text{tg}(5x)\,dx$

se $u = \text{tg}(5x)$
$du = \sec^2(5x)\,dx$
$\int u\,du = \frac{u^2}{2}$

$u = 5x \quad du = 5dx \quad \frac{du}{5} = dx$
$\int \text{tg}\,u\,\frac{du}{5} = \frac{1}{5}\cdot\ln|\sec u|$

$\boxed{\int \text{tg}^3(x)\,dx = \frac{\text{tg}^2(5x)}{2} - \frac{1}{5}\ln|\sec(5x)| + c}$

EXERCÍCIOS - CALCULE AS INTEGRAIS

EX01 $\int \text{sen}^5(3x)\,dx$

EX02 $\int \cos^4(2x)\,dx$

EX03 $\int \text{sen}^2(x)\cdot\cos^4(x)\,dx$

EX04 $\int \text{sen}(4x)\cdot\cos(2x)\,dx$

EX05 $\int \sec^3(x)\,dx$

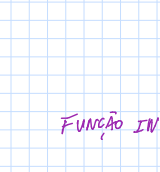
EX06 $\int \text{cotg}^2(6x)\,dx$

PÁG 261

INTEGRAIS POR SUBSTITUIÇÃO TRIGONOMÉTRICA

$a \in \mathbb{R}$
$u \rightarrow$ variável

FUNÇÃO INTEGRANDO ENVOLVE $\sqrt{a^2 - u^2}$



Substituição $u = a\,\text{sen}\,\theta$
$du = a.\cos\theta\,d\theta$ $\quad \forall\, -\pi/2 \leq \theta \leq \pi/2$

$\sqrt{a^2 - u^2} = \sqrt{a^2 - a^2\text{sen}^2\theta}$
$= \sqrt{a^2(1 - \text{sen}^2\theta)}$
$= a.\sqrt{1 - \text{sen}^2\theta}$
$= a.\sqrt{\cos^2\theta}$

$\boxed{\text{sen}^2\theta + \cos^2\theta = 1}$

$\sqrt{a^2 - u^2} = a.\cos\theta$

EX01 $\int \frac{\sqrt{16 - x^2}}{x^2}\,dx$ $\quad \sqrt{a^2 - u^2}$; $a^2 = 16 \quad a = 4$; $u^2 = x^2 \quad u = x$

$x = 4\,\text{sen}\,\theta$
$dx = 4.\cos\theta\,d\theta$
$\sqrt{16 - x^2} = 4\cos\theta$

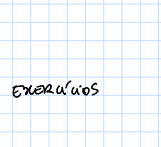
$\int \frac{\sqrt{16 - x^2}}{x^2}\,dx = \int \frac{4\cos\theta.\,4\cos\theta\,d\theta}{(4\,\text{sen}\,\theta)^2}$
$= \int \frac{16\cos^2\theta}{16\,\text{sen}^2\theta}\,d\theta$
$= \int \text{cotg}^2\theta\,d\theta$
$= \int (\text{cossec}^2\theta - 1)\,d\theta$
$= \int \text{cossec}^2\theta\,d\theta + \int -1\,d\theta$
$= -\text{cotg}\,\theta - \theta + c$
$= -\frac{\sqrt{16 - x^2}}{x} - \text{arc.sen}\left(\frac{x}{4}\right) + c$

$\text{cotg}^2\theta = \text{cossec}^2\theta - 1$

$\text{cotg}\,\theta = \frac{ca}{co} \rightarrow \text{cotg}\,\theta = \frac{\sqrt{16 - x^2}}{x}$

$x = 4.\text{sen}\,\theta \rightarrow \frac{x}{4} = \text{sen}\,\theta \rightarrow \theta = \text{arc sen}\left(\frac{x}{4}\right)$

FUNÇÃO INTEGRANDO ENVOLVE $\sqrt{u^2 + a^2}$



Substituindo $u = a.\text{tg}\,\theta$
$du = a.\sec^2\theta\,d\theta$ $\quad \forall\, -\pi/2 \leq \theta \leq \pi/2$

$\sqrt{u^2 + a^2} = \sqrt{a^2 + a^2\text{tg}^2\theta}$
$= \sqrt{a^2(1 + \text{tg}^2\theta)}$
$= a.\sqrt{1 + \text{tg}^2\theta}$
$= a.\sqrt{\sec^2\theta}$
$= a.\sec\theta$

$\boxed{\sec^2\theta = 1 + \text{tg}^2\theta}$

EX02 $\int \sqrt{x^2 + 5}\,dx$ $\quad \sqrt{u^2 + a^2}$; $a^2 = 5 \rightarrow a = \sqrt{5}$; $u^2 = x^2 \rightarrow x = u$

$x = \sqrt{5}\,\text{tg}\,\theta$
$dx = \sqrt{5}.\sec^2\theta\,d\theta$
$\sqrt{x^2 + 5} = \sqrt{5}.\sec\theta$

$\int \sqrt{x^2 + 5}\,dx = \int \sqrt{5}.\sec\theta.\sqrt{5}\sec^2\theta\,d\theta$
$= 5\int \sec\theta.\sec^2\theta\,d\theta$

$u = \sec\theta \quad du = \sec\theta.\text{tg}\,\theta\,d\theta$
$\int dv = \int \sec^2\theta\,d\theta \quad v = \text{tg}\,\theta$

$\int u\,dv = uv - \int v\,du$ "INTEGRAÇÃO POR PARTES"

$\int \sec\theta.\sec^2\theta\,d\theta = \sec\theta.\text{tg}\,\theta - \int \text{tg}\,\theta.\sec\theta.\text{tg}\,\theta\,d\theta$
$= \sec\theta.\text{tg}\,\theta - \int \sec\theta.\text{tg}^2\theta\,d\theta$
$= \sec\theta.\text{tg}\,\theta - \int \sec\theta.(\sec^2\theta - 1)\,d\theta$
$= \sec\theta.\text{tg}\,\theta - \int \sec^3\theta - \sec\theta\,d\theta$

$\boxed{\text{tg}^2\theta = \sec^2\theta - 1}$

$\int \sec^3\theta\,d\theta = \sec\theta.\text{tg}\,\theta - \int \sec^3\theta\,d\theta + \int \sec\theta\,d\theta$

$2\int \sec^3\theta\,d\theta = \sec\theta.\text{tg}\,\theta + \ln|\sec\theta + \text{tg}\,\theta| + c$

$\int \sec^3\theta\,d\theta = \frac{1}{2}.\left[\sec\theta.\text{tg}\,\theta + \ln|\sec\theta + \text{tg}\,\theta| + c\right]$

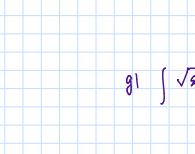
$\int \sqrt{x^2 + 5}\,dx = \frac{1}{2}.\left(\frac{\sqrt{x^2 + 5}}{\sqrt{5}}.\frac{x}{\sqrt{5}} + \ln\left|\frac{\sqrt{x^2 + 5}}{\sqrt{5}} + \frac{x}{\sqrt{5}}\right| + c\right)$

$\int \sqrt{x^2 + 5}\,dx = \frac{x.\sqrt{x^2 + 5}}{10} + \frac{\ln}{2}\left|\frac{\sqrt{x^2 + 5}}{\sqrt{5}} + \frac{x}{\sqrt{5}}\right| + c$

$\sec\theta = \frac{h}{ca} \Rightarrow \sec\theta = \frac{\sqrt{x^2 + 5}}{\sqrt{5}}$

$\text{tg}\,\theta = \frac{co}{ca} \Rightarrow \text{tg}\,\theta = \frac{x}{\sqrt{5}}$

FUNÇÃO INTEGRANDO ENVOLVE $\sqrt{u^2 - a^2}$



Substituindo $u = a.\sec\theta$
$du = a.\sec\theta.\text{tg}\,\theta\,d\theta$ $\quad \forall\, 0 \leq \theta < \pi/2$ ou $\pi \leq \theta < 3\pi/2$

$\sqrt{u^2 - a^2} = \sqrt{a^2\sec^2\theta - a^2}$
$= \sqrt{a^2(\sec^2\theta - 1)}$
$= a.\sqrt{\text{tg}^2\theta}$

$\sqrt{u^2 - a^2} = a.\text{tg}\,\theta$

EX03 $\int \frac{dx}{\sqrt{x^2 - 25}}$ $\quad \sqrt{x^2 - 25} = \sqrt{u^2 - a^2}$; $a^2 = 25 \rightarrow a = 5$; $x^2 = u^2 \rightarrow x = u$

$x = 5.\sec\theta$
$dx = 5.\sec\theta.\text{tg}\,\theta\,d\theta$
$\sqrt{x^2 - 25} = 5\,\text{tg}\,\theta$

$\int \frac{dx}{\sqrt{x^2 - 25}} = \int \frac{5\sec\theta.\text{tg}\,\theta\,d\theta}{5\,\text{tg}\,\theta}$
$= \int \sec\theta\,d\theta$
$= \ln|\sec\theta + \text{tg}\,\theta| + c$
$= \ln\left|\frac{x}{5} + \frac{\sqrt{x^2 - 25}}{5}\right| + c$
$= \ln\left|\frac{x + \sqrt{x^2 - 25}}{\ln(5)}\right| + c$
$= \ln|x + \sqrt{x^2 - 25}| - \underbrace{\ln(5) + c}_{c_2}$

$\sec\theta = \frac{h}{ca} = \frac{x}{5}$

$\text{tg}\,\theta = \frac{co}{ca} = \frac{\sqrt{x^2 - 25}}{5}$

$\int \frac{dx}{\sqrt{x^2 - 25}} = \ln|x + \sqrt{x^2 - 25}| + c_2$

EXERCÍCIOS

EX01 $\int \frac{dx}{x\sqrt{x^2 + 4}}$

EX02 $\int \frac{\sqrt{9 - x^2}}{x^2}\,dx$

EX03 $\int \frac{dx}{x\sqrt{25 - x^2}}$

EX04 $\int \frac{dx}{x^2\sqrt{x^2 - 7}}$

EX05 $\int \frac{dx}{x^2.\sqrt{7 - x^2}}$

EXERCÍCIO 6.9 PÁG 273

EX01 a; b; c; e; f; g, K; m; h;

d) $\int \frac{\ln^3(x)}{x}\,dx = \int (\ln x)^3\,\frac{dx}{x} \longrightarrow \int u^3\,du = \frac{u^4}{4} + c = \frac{\ln^4(x)}{4} + c$

$u = \ln(x)$
$\frac{du}{dx} = \frac{1}{x} \rightarrow du = \frac{1dx}{x}$

g) $\int \sqrt{\text{sen}(x)}.\cos(x)\,dx \longrightarrow \int \sqrt{u}\,du = \int u^{1/2}\,du = \frac{u^{3/2}}{3/2} + c = \frac{2}{3}\,\text{sen}^{3/2}(x) + c$ $\quad (\text{sen}(x))^{3/2}$

$u = \text{sen}(x)$
$\frac{du}{dx} = \cos(x) \rightarrow du = \cos(x)dx$

K) $\int \frac{e^{1/x} + 1}{x^2}\,dx \longrightarrow \int e^u + 1\,(-du) = -\int (e^u + 1)\,du$
$= -e^u + u + c$
$= -e^{1/x} + \frac{1}{x} + c$

$u = \frac{1}{x} \rightarrow u = x^{-1}$
$\frac{du}{dx} = -1x^{-2}$
$du = -\frac{1}{x^2}dx$
$-du = \frac{dx}{x^2}$

EX02 a, b, c, e, f, h.

EX03 a, b, e, f, g

EX04 a, b, h.

EX06 a, b, c, d, e, h

EX08 g, h, i, j, K, l, s, u, ...

$\int \frac{\ln x^2}{x}\,dx$

$u = \ln x^2$
$\frac{du}{dx} = \frac{2x}{x^2} \rightarrow du = \frac{2}{x}dx \rightarrow \frac{du}{2} = \frac{dx}{x}$

$\int \frac{\ln x^2}{x}\,dx = \int u\,\frac{du}{2} = \frac{1}{2}\frac{u^2}{2} + c$
$= \frac{1}{4}.(\ln x^2)^2 + c$

$f(x) = \ln(x) \quad y = \ln(u) \quad y' = \frac{u'}{u}$

$f'(x) = \frac{1}{x}$

$\int f(x)\,dx = \int \underbrace{\ln(x)}_{u}\,\overbrace{dx}^{dv}$

$u = \ln x \quad du = \frac{1}{x}dx$
$\int dv = \int dx \quad v = x$

$= \ln(x).x - \int x.\frac{1}{x}\,dx$
$= x.\ln(x) - \int dx$
$= x\ln(x) - x + c$

$\int x.\sqrt{x + 1}\,dx \xrightarrow{\text{INT. PARTES}} u.v - \int v\,du$

$u = x \quad du = 1dx$
$\int dv = \int (x + 1)^{1/2}\,dx \quad v = \frac{2}{3}(x + 1)^{3/2}$

$\int (x + 1)^{1/2}\,dx$; $u = x + 1$, $du = 1dx$; $\int u^{1/2}\,du = \frac{u^{3/2}}{3/2} = \frac{2}{3}(x + 1)^{3/2}$

$\int x\sqrt{x + 1}\,dx = x.\frac{2}{3}(x + 1)^{3/2} - \int \frac{2}{3}(x + 1)^{3/2}\,dx$
$= \frac{2}{3}x(x + 1)^{3/2} - \frac{2}{3}\int (x + 1)^{3/2}\,dx$
$= \frac{2}{3}x(x + 1)^{3/2} - \frac{2}{3}.\frac{2}{5}(x + 1)^{5/2} + c$
$= \frac{2}{3}x(x + 1)^{3/2} - \frac{4}{15}(x + 1)^{5/2} + c$

$u = x + 1$; $du = 1dx$; $\int u^{3/2}\,du = \frac{u^{5/2}}{5/2} = \frac{2}{5}u^{5/2}$

EX06 $\int \frac{e^x\,dx}{e^{2x} + 16} = \int \frac{e^x\,dx}{(e^x)^2 + 4^2}$
$= \int \frac{du}{u^2 + 4^2}$
$= \frac{1}{4}.\text{arc tg}\,\frac{u}{4} + c$
$= \frac{1}{4}\,\text{arc tg}\,\frac{e^x}{4} + c$

$u = e^x$
$du = e^x\,dx$

EX05 $\int \frac{\ln x^2}{x}\,dx$

$u = \ln x^2$
$du = \frac{2x}{x^2}\,dx \rightarrow du = \frac{2}{x}dx$
$\frac{du}{2} = \frac{dx}{x}$

$= \int u\,\frac{du}{2}$
⋮

EX03 $\int \frac{1}{x^3 + 8}\,dx$

$x^3 + 8 = 0$
$x^3 = -8$
$x = \sqrt[3]{-8}$
$x = -2 \rightarrow (x + 2)$

$x^3 + 8 \,|\, x + 2$
$-x^3 - 2x^2 \quad x^2 - 2x + 4$
$-2x^2 + 8$
$+2x^2 + 4x$
$4x + 8$
$-4x - 8$
0

$x^3 + 8 = (x + 2).(x^2 - 2x + 4)$
LINEAR; $x^2 - 2x + 4 = 0$ QUADRÁTICO IRREDUTÍVEL; $\frac{2 \pm \sqrt{4 - 16}}{2}$

$\frac{1}{x^3 + 8} = \frac{A}{x + 2} + \frac{Bx + C}{x^2 - 2x + 4}$

$\frac{1}{x^3 + 8} = \frac{A(x^2 - 2x + 4) + (Bx + C).(x + 2)}{(x + 2).(x^2 - 2x + 4)}$

$1 = Ax^2 - 2Ax + 4A + Bx^2 + 2Bx + Cx + 2C$

$\begin{cases} A + B = 0 \\ -2A + 2B + C = 0 \\ 4A + 2C = 1 \end{cases}$

$\rightarrow A = -B \rightarrow -A = B$

$\rightarrow 2C = 1 - 4A$
$C = \frac{1 - 4A}{2}$

$-2A + 2.(-A) + \frac{1 - 4A}{2} = 0$
$-2A - 2A + \frac{1 - 4A}{2} = 0$
$\frac{-8A + 1 - 4A}{2} = 0$
$-12A = -1$
$\boxed{A = 1/12}$
$\boxed{B = -1/12}$

$C = \frac{1 - 4(1/12)}{2}$
$C = \frac{1 - 1/3}{2}$
$C = \frac{2/3}{2} \rightarrow C = \frac{2}{3}.\frac{1}{2} = \frac{1}{3}$

$\frac{1}{x^3 + 8} = \frac{1/12}{x + 2} + \frac{-1/12\,x + 1/3}{x^2 - 2x + 4}$

$-\frac{1}{12}x + \frac{1}{3} = -\frac{1}{12}.(x - 4)$

$\int \frac{1}{x^3 + 8}\,dx = \int \frac{1/12}{x + 2}\,dx + \int \frac{-1/12\,x + 1/3}{x^2 - 2x + 4}\,dx$
$= \frac{1}{12}\int \frac{dx}{x + 2} - \frac{1}{12}\int \frac{(x - 4)\,dx}{x^2 - 2x + 4} \rightarrow x^2 - 2x + 1 + 3 = (x - 1)^2 + 3$
$= \frac{1}{12}\int \frac{dx}{x + 2} - \frac{1}{12}\left[\int \frac{(x - 1) - 3}{(x - 1)^2 + 3}\,dx\right]$
$= \frac{1}{12}\int \frac{dx}{x + 2} - \frac{1}{12}\left[\int \frac{x - 1}{(x - 1)^2 + 3}\,dx + \int \frac{-3}{(x - 1)^2 + 3}\,dx\right]$
$= \frac{1}{12}\int \frac{du}{u} - \frac{1}{12}\left[\int \frac{u\,du}{u^2 + 3} - 3\int \frac{du}{u^2 + 3}\right]$

$u = x + 2 \quad du = dx$; $u = x - 1 \quad du = dx$; $u = x - 1 \quad du = dx$